Human Factors & Measurement

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Lesson outline

- Fundamentals of psychometrics
- Usability scales
- Mental workload
- Data quality in empirical studies

Human Factors

Human factors (or ergonomics): understanding human capabilities and limits (cognitive, physical, social) and designing systems around them.

In **HCI**, human factors is about how people **perceive**, **think**, **move**, **decide**, and **fail** when interacting with digital systems.

Measurement is central because:

- We want to know how well a system fits the human (performance, errors, workload, satisfaction).
- **We need numbers** (read “facts and data”) to justify design decisions and claims.

Notice how they map to ISO and Nielsen definitions of usability.

Why Measure in HCI?

Without measurement, design/research is mostly opinion, habit, or aesthetics.

We want evidence about:

- **Performance**: task time, error rate, throughput, learning curve.
- **Experience**: perceived usability, mental workload, reliance, perceived control.
- **Safety**: near-misses, critical incidents, recovery.

By measuring we are allowed to:

- **Compare** versions, devices, or interaction techniques.
- **Diagnose** where and why users struggle.
- Build cumulative **know**ledge: results that generalize beyond one project.

From Constructs to Indicators

Most things we care about are **latent constructs**: “usability”, “mental workload”, “reliance”, “engagement”, “learning”.

We don’t observe them directly, **we observe indicators**: questionnaires, task times, error counts, clicks, eye movements, physiology.

Mapping problem:

- ➔ Same construct can have multiple indicators.
- ➔ Same indicator can be influenced by different constructs (long task time = bad UI, or just unfamiliarity).

Good measurement requires:

- A clear definition of the construct.
- A plausible link between construct and indicator.
- Awareness of limitations and alternative explanations.

Fundamentals of Psychometrics

IUM 25-26

Psychometrics in a Nutshell

Psychometrics is the science of **designing**, **evaluating**, and **interpreting psychological measures**.

Main goals:

- Create instruments that are reliable (stable, consistent).
- Ensure they are valid (actually measure the intended construct).
- Understand measurement error and its impact on conclusions.

In HCI we borrow and adapt scales (SUS, NASA-TLX, reliancscales, etc.).

We also sometimes build new scales for new technologies or contexts (CUS).

Measuring the Right Construct

Validity: are we measuring what we think we are measuring?

Main aspects:

- **Content** validity: items adequately cover the construct's domain (usability = effectiveness, efficiency, satisfaction).
- **Construct** validity:
  - **Convergent**: correlates with related constructs (e.g., two usability scales).
  - **Discriminant**: doesn't correlate too strongly with unrelated constructs (e.g., usability vs personality).
- **Criterion** validity: predicts relevant outcomes (higher usability scores → fewer errors, faster task times).

A scale can be highly reliable but invalid (measures something else very consistently).

Popularity is not proof of validity in your specific context of use.

Consistency of Measurement

Reliability ≈ how much of the score is signal vs noise.

3 types:

- **Internal consistency**: do items that are supposed to measure the same thing correlate? (Cronbach's alpha, omega).
- **Test–retest reliability**: if the underlying construct doesn't change, do scores remain similar over time?
- **Inter-rater reliability**: when human raters code behavior, do they agree?

Low reliability → more random error:

1. Reduces statistical power (harder to detect real effects).
2. Can mask real differences between interfaces.

Practical actions:

- Check item–total correlations; remove or revise problematic items.
- Avoid extremely vague or double-barreled items.

Measurement Scales & Likert Responses

Quantitative variables differ in scale type:

- **Nominal**: categories without order (device: phone, tablet, desktop).
- **Ordinal**: ordered but unknown interval size (Likert “strongly disagree”→“strongly agree”).
- **Interval**: equal distances, but arbitrary zero (°C, calendar years) → you **CANNOT** multiply!
- **Ratio**: equal distances, meaningful zero (duration in s, Kelvin, euros) → you **CAN** multiply!

Likert items:

- Strictly ordinal, but often treated as approx. interval when:
- We aggregate several items into sum or mean scores.
- Response options are balanced and symmetric.

<u>This is extremely important when considering which the hypothesis test to employ!</u>

Usability Scales

Measuring Usability as a Construct

Usability (ISO 9241-11)

The degree to which a system can be used by specified users to achieve specified goals **effectively, efficiently, and satisfactorily** in a specified context of use.

Questionnaires are quick to administer, allow users to summarize their experience, and provide a single or few numbers to compare systems.

But perceived usability can diverge from objective performance (people can like a system that is actually slower).

Good practice:

- Combine subjective usability scores with task performance.
- Select scales that match your system and research question.

System Usability Scale (SUS)

- ➔ 10 items, 5-point agreement scale, technology-agnostic.
- ➔ The odd-numbered questions (1, 3, 5, 7, 9) are positive statements, the even-numbered questions (2, 4, 6, 8, 10) are negative statements.
- ➔ Items are summed → 0-100 score.

Strengths:

- Huge body of data and meta-analyses in the literature.
- Works across many domains, quick to administer.

Limitations:

- Some items are awkwardly worded or dated.
- Mixes usability with attitudinal acceptance and perceived satisfaction.

Use when:

1. You need a **simple**, **comparable** global **usability score**.
2. You want to **benchmark against other systems** or prior studies.

UMUX and UMUX-Lite

- ➔ UMUX (Usability Metric for User Experience): 4 items focused on ISO usability components (usefulness, ease of use, etc.).
- ➔ UMUX-Lite: 2 items (perceived usefulness + ease of use); extremely short.

Strengths:

- Less burden compared to SUS.
- Often correlates strongly with SUS.

Limitations:

- Fewer items → more sensitive to noise or context.
- Harder to diagnose why usability is low.

Use when:

1. Survey **time is scarce** (mobile, in-the-wild studies).
2. You need a global usability measurement but **can't afford 10 items** (participants' attention, many post task surveys)

Chatbot Usability Scale (BUS-11)

- ➔ Dedicated scale for chatbot-specific usability / satisfaction, capturing things generic scales (SUS, UMUX) miss: conversation quality, context management, privacy, response time, not just “ease of use”.
- ➔ **BUS-11**: 11 items, 5-point Likert
- ➔ Provides also an heuristic approach for designers (BUS-CHECK)

Strengths:

- Specific for chatbots.
- Validated across several languages and contexts (EN, DE, NL, ES, IT), suitable for cross-country studies
- Strong convergent validity with UMUX-Lite

Limitations:

- Relatively young, evidence in literature is still thinner.
- Mainly focused on satisfaction and perceived quality.

Use when:

1. You want chatbot-specific insight **beyond SUS/UMUX**.
2. You **compare** multiple **chatbots** or versions, also in **different languages**.

Mental Workload

Why We Care About It?

Mental workload can be roughly defined as the **amount of cognitive resources** required **to meet task demands**.

Usually, it depends on:

- Task complexity and time pressure.
- Interface/technology (clarity, feedback, information density).
- User factors (expertise, fatigue, stress).

Theoretically the expected consequences should:

- **Too low** → very easy tasks, higher performance, mostly automatic.
- **Too high** → errors, slower performance, unsafe behavior, stress.

But this is not always the case, or it could be even wrong!

Not Gold pt.1

There are **tens of possible definitions** of Mental Workload.

Before using it, **clarify if it make sense and how** in your study.

And remember, **Mental Workload ≠ Cognitive Load**.

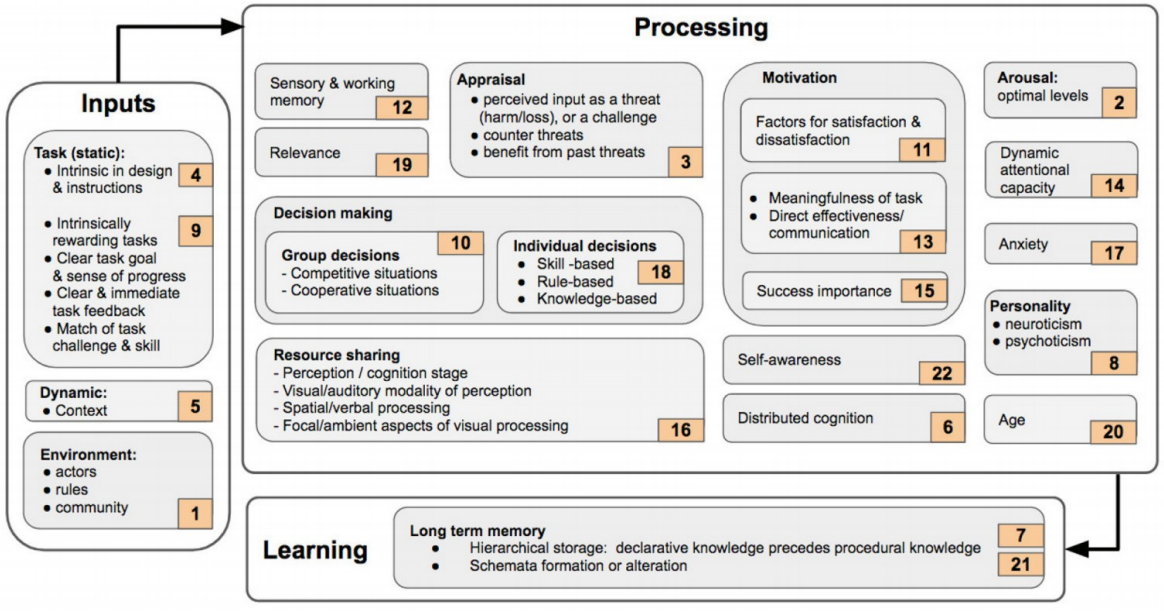
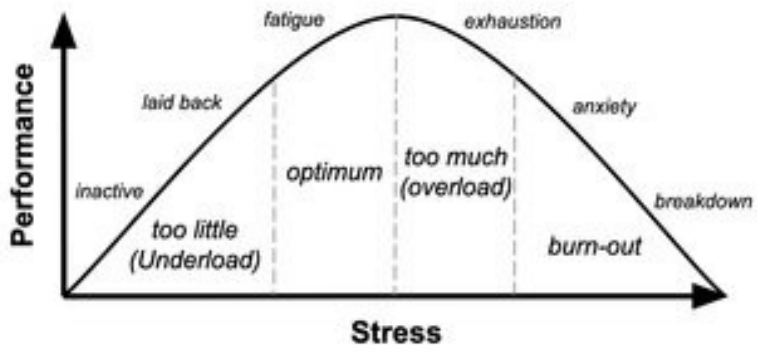


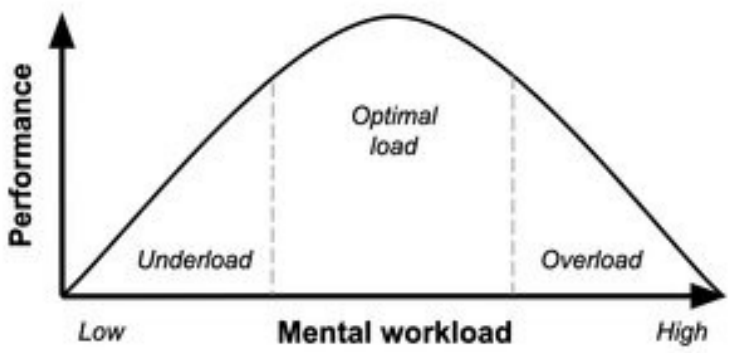
FIGURE 8 | Theories linked to the construct of mental workload organized as a framework.

L. Longo, C. D. Wickens, G. Hancock, and P. A. Hancock, “Human Mental Workload: A Survey and a Novel Inclusive Definition,” Front. Psychol., vol. 13, June 2022, doi: 10.3389/fpsyg.2022.883321.

ALl that Glitters Is Not Gold pt.2

Moreover, the **relationship** between mental **workload** and **performance** is **unclear**: it could be that it follows the **Yerkes-Dodson Law**.





L. Longo and M. Rajendran, “A Novel Parabolic Model of Instructional Efficiency Grounded on Ideal Mental Workload and Performance,” in Human Mental Workload: Models and Applications, L. Longo and M. C. Leva, Eds., Cham: Springer International Publishing, 2021, pp. 11–36. doi: 10.1007/978-3-030-91408-0_2.

MWL Measurement: NASA-TLX

- **Six levels**: mental, physical, temporal demand, performance, effort, frustration.
- 2 variants:
  - Original includes **weighting** between dimensions (pairwise comparisons).
  - **Raw TLX** uses **unweighted** average of ratings (simpler, widely used, equally valid).

Advantages:

- **Quick** to administer (only 6 questions).
- **Very common** in human factors, aviation, automotive, healthcare, HCI.

Limitations:

- It is **subjective**.
- **Memory-based**: cannot capture workload in real time.
- It is **not linear**: the steps are not equal → you cannot say "*Condition A has 30% more workload than Condition B*"*

See it as a **comparative tool between conditions** (interfaces, tasks), not as an absolute truth about "true workload".

* On limitations, see: E. Babaei, T. Dingler, B. Tag, and E. Velloso, "Should we use the NASA-TLX in HCI? A review of theoretical and methodological issues around Mental Workload Measurement," International Journal of Human-Computer Studies, vol. 201, p. 103515, June 2025, doi: 10.1016/j.ijhcs.2025.103515.

NASA-TLX

Mental Demand — How mentally demanding was the task?

Very Low — Very High

Physical Demand — How physically demanding was the task?

Very Low — Very High

Temporal Demand — How hurried or rushed was the pace of the task?

Very Low — Very High

Performance — How successful were you in accomplishing what you were asked to do?

Perfect — Failure

Effort — How hard did you have to work to accomplish your level of performance?

Very Low — Very High

Frustration — How insecure, discouraged, irritated, stressed, and annoyed wereyou?

Very Low — Very High

Other Subjective Workload Measures

- **SWAT (Subjective Workload Assessment Technique)**: asks about time pressure, mental effort, and stress, then the overall is converted to %.
- **RSME** (Rating Scale Mental Effort): single continuous scale for mental effort.
- **Bedford Workload Scale**: unidimensional 10-point rating scale, structured as a hierarchical decision tree.
- **MRQ (Multiple Resources Questionnaire)**: 17-item questionnaire based on Multiple Resource Theory (visual vs auditory, spatial vs verbal, etc.)

Trade-offs:

- **Short scales** (1–3 items)
  - + Very fast, low burden.
  - – Less detailed, mainly good for simple comparisons (“A > B”).
- **Longer** / multi-dimensional **scales**
  - + More diagnostic: show why workload is high (time pressure? info overload? stress?).
  - – Take longer to fill in, can cause fatigue if repeated often.

Physiological Indicators of Workload

- Heart rate (HR) and heart rate variability (HRV) → arousal, stress, effort.
- Pupil dilation → cognitive load, but sensitive to lighting.
- Skin conductance (EDA/GSR) → sympathetic arousal.
- Eye tracking → fixation patterns, blink rate, gaze transitions.
- EEG / fNIRS* → brain-level signals.

Pros:

- **Continuous over time**: can capture dynamic changes within a task.
- Less prone to self-report biases.

Cons:

- **Intrusive** or uncomfortable sensors, sensitive to artifacts (movement, environment).
- **Interpretation is indirect**: many signals correlate with general arousal, not only workload.
- Often require a **lab**.

Good practice:

- Probably it is a good idea to use physiology to complement behavioral and subjective data, not to replace them.

* EEG: electroencephalogram,
fNIRS: functional near-infrared spectroscopy

Data Quality

Data Quality as a Research Problem

Poor data quality = poor measurement of humans

Main threats:

- Inattentive or random responding (speeding, straight-lining).
- Misunderstood tasks or questions (ambiguous wording, poor examples).
- Demand characteristics: participants respond according to what they think you want. Really insidious!
- Technical issues: logging errors, missing data, timeouts.

In online HCI studies, participants multi-task, use small screens, get distracted, or run ad-blockers that break logging.

Result:

Biased or noisy data → **false negatives** (missing real effects) or **false positives** (seeing effects that aren't robust).

Designing for Better Data Quality

Study design:

- **Clear**, **concise instructions** with examples and maybe a practice block.
- **Limit** total session **duration**.
- Align **compensation** with effort.

Instruments:

- **Pilot** test **questionnaires** and **tasks** with a small sample.
- **Avoid jargon** and **multi-barreled questions** (asking more than a topic at once).

Checks and filters:

- **Attention checks** (used carefully and transparently).
- **Comprehension checks** to test whether a participant has understood critical information.
- **Look** out for **bots**, **cheaters** and **slackers**:
  - Monitor task **completion times**: too fast or too slow observations may be suspect.
  - Look for **patterns**: straight-lining, impossible answer combinations.